

On the Inevitability of 4-Cycles in Tanner Graphs of CSS Codes

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In this document, we comment on the inevitability of 4-Cycles in Tanner Graphs of CSS codes as follows:

It should be noted that even if the individual Tanner graphs of the classical codes used in CSS constructions are free of 4-cycles, the joint Tanner graph of the resulting quantum code necessarily contains 4-cycles. The reader should note that the joint Tanner graph is formed by combining the Tanner graphs of both underlying classical codes from X and Z stabilizer matrices.

Now, we support our claim that the **joint** Tanner graphs of CSS (unassisted) codes necessarily contain 4-cycles, as explained below:

Construction of CSS code from two different binary classical linear codes

Consider the following two different parity check matrices for the binary linear codes C_1 and C_2

$$H_1 = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, H_2 = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{l1} & b_{l2} & \cdots & b_{ln} \end{bmatrix}, \quad (1)$$

respectively, where $a_{i,j}, b_{k,j} \in \mathbb{F}_2$.

Assume that neither H_1 nor H_2 contains any zero rows or zero columns. The (i, j) -th entry of $H_1 H_2^T$, where H_2^T denotes the transpose of H_2 , is given by

$$\sum_{k=1}^n a_{ik} b_{jk}.$$

In order to satisfy the orthogonality condition, we must have

$$\sum_{k=1}^n a_{ik} b_{jk} = 0 \pmod{2} \quad \text{for all } 1 \leq i \leq m, 1 \leq j \leq l, \quad (2)$$

which implies that this sum must be an even integer. Under the assumption that all row and column weights are nonzero, it follows from Eq. (2) that

$$\sum_{k=1}^n a_{ik} b_{jk} \geq 2 \text{ (even)}, \quad (3)$$

which leads to the existence of 4-cycles in the joint Tanner graph of the resulting CSS code, as the joint Tanner is constructed by combining the matrices H_1 and H_2 as follows:

$$H = \begin{bmatrix} H_1 \\ H_2 \end{bmatrix}.$$

The joint Tanner graph induced by the matrix H is free of 4-cycles if the pairwise inner product between any two rows of H is at most 1, as characterized in Eq. 3

Construction of a CSS code from one binary classical linear code

Let

$$H = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be a parity-check matrix of a binary linear code C , where $m < n$. The (i, j) -th entry of HH^T is given by

$$\sum_{k=1}^n a_{ik}a_{jk},$$

which is the (non-mod-2) inner product of the i -th and j -th rows of H .

It is well-known that a 4-cycle exists in the Tanner graph of H if and only if the inner product (non-mod-2) of two distinct rows of H is at least 2. On the other hand, to satisfy the condition $C^\perp \subseteq C$, we must have

$$HH^T = 0 \pmod{2},$$

which implies that $\sum_{k=1}^n a_{ik}a_{jk}$ is an even integer for all i, j .

For $i = j$, this condition yields

$$\sum_{k=1}^n a_{ik}^2 = \sum_{k=1}^n a_{ik} = 0 \pmod{2},$$

which implies that each row of H must have even weight, and in particular weight at least 2 (otherwise the row would be identically zero).

If we further assume that each column of H has weight at least 2, as is typical in LDPC constructions, then there exist indices $i_1 \neq j_1$ such that

$$\sum_{k=1}^n a_{i_1 k}a_{j_1 k} \geq 2,$$

which necessarily induces a 4-cycle in the Tanner graph of the code defined by H . Consequently, under standard LDPC constraints, the presence of short cycles in the Tanner graph of H is unavoidable, and therefore such cycles are also unavoidable in the joint Tanner graph of the corresponding CSS code.

The reader should note that when a CSS code is constructed from a single classical code, neither the underlying classical code nor the joint Tanner graph of the resulting CSS

code can avoid 4-cycles. In contrast, in the entanglement-assisted setting, the underlying classical code may be free of 4-cycles in its Tanner graph; however, the unassisted portion of the joint Tanner graph of the EA-CSS code still contains 4-cycles.

However, by relaxing the column-weight constraint, one can construct a CSS code whose underlying classical code is free of short cycles; however, such constructions are typically inefficient for communication purposes. For instance, consider the following parity-check matrix:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The Tanner graph corresponding to this matrix contains no short cycles and satisfies the orthogonality condition $HH^T = 0 \pmod{2}$.

In contrast, for entanglement-assisted quantum codes, the entanglement-unassisted portion of the joint Tanner graph can be made free of 4-cycles. In this setting, the unassisted portion of Tanner graph is obtained by removing the variable nodes corresponding to the receiver-side ebits, as these ebits are assumed to be error-free and do not participate in the decoding process. Since the receiver-side ebits are removed from the joint Tanner graph in the entanglement-assisted case so it breaks the CSS-structure over the joint Tanner graph and make it 4-cycle free. The reader is referred to Fig. 1 for an illustration of the unassisted portion of the joint Tanner graph.

Finally, the following example demonstrates that 4-cycles can be circumvented within the entanglement-assisted framework.

Example 0.1 Let C_1 and C_2 be two $[9, 6]_2$ classical QC-LDPC codes with the following parity-check matrices:

$$H_x = \left[\begin{array}{ccc|ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \end{array} \right],$$

$$H_z = \left[\begin{array}{ccc|ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \end{array} \right],$$

respectively. Then by using C_1 and C_2 an EA-QC-QLDPC code with parameters $[[9, 4, 2; 1]]_2$ can be constructed, where the extended parity-check matrices are:

$$\left. \begin{aligned} H_{ex} &= \left[\begin{array}{ccc|ccc|ccc||c} 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right], \\ H_{ez} &= \left[\begin{array}{ccc|ccc|ccc||c} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \end{array} \right]. \end{aligned} \right\}$$

The Tanner graph of the EA-QC-QLDPC code is shown in Fig. 1. It clearly demonstrates that the entanglement-unassisted portion of the Tanner graph is free of 4-cycles.

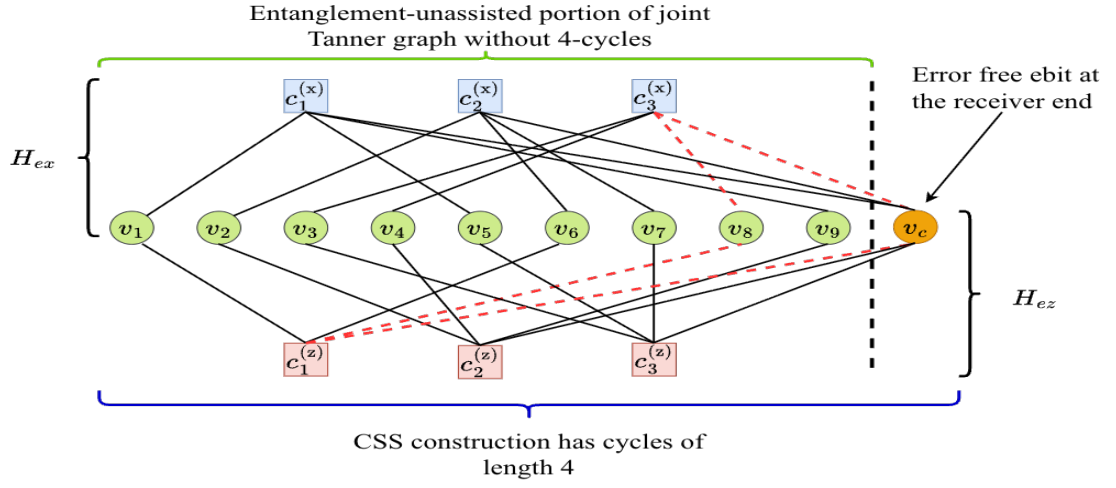


Figure 1: Tanner graph for the code $[[9, 4, 2; 1]]_2$ contains nine variable nodes $v_1, \dots, v_9$ at the transmitter end and variable node v_c at the receiver end corresponding to the pre-shared entangled bits. Since the node v_c is assumed to be error-free, it does not participate in the decoding. For entanglement-unassisted case, the Tanner graph has short cycles of length 4, as shown by red dotted lines. This is due to the CSS construction.